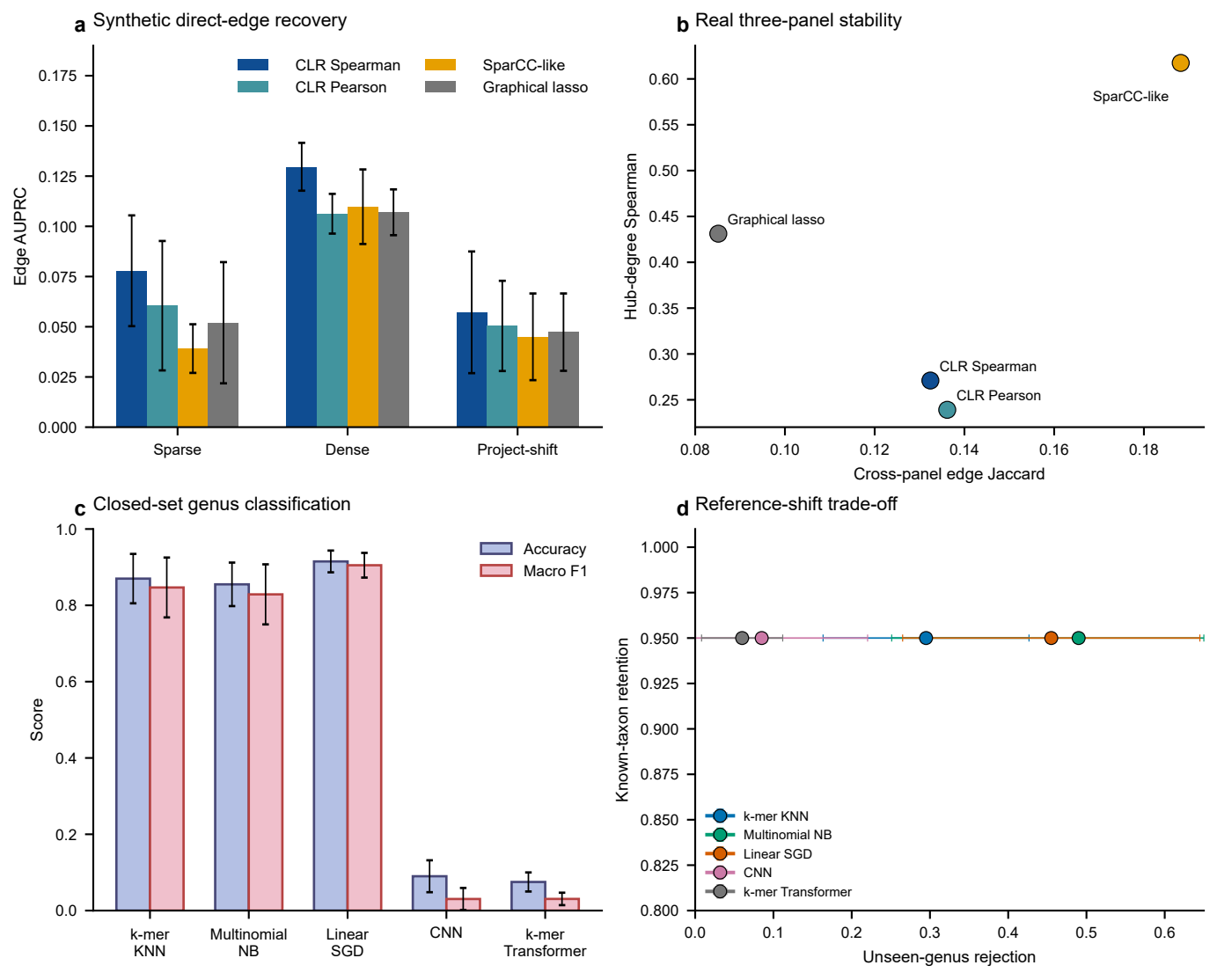
a
Synthetic direct-edge recovery
CLR Spearman
CLR Pearson
SparCC-like
Graphical lasso
Edge AUPRC
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
Sparse
Dense
Project-shift
b
Real three-panel stability
Hub-degree Spearman
0.60
0.55
0.50
0.45
0.40
0.35
0.30
0.25
SparCC-like
Graphical lasso
CLR Spearman
CLR Pearson
0.08
0.10
0.12
0.14
0.16
0.18
Cross-panel edge Jaccard
c
Closed-set genus classification
Accuracy
Macro F1
Score
1.0
0.8
0.6
0.4
0.2
0.0
k-mer KNN
Multinomial NB
Linear SGD
CNN
k-mer Transformer
d
Reference-shift trade-off
Known-taxon retention
1.000
0.975
0.950
0.925
0.900
0.875
0.850
0.825
0.800
k-mer KNN
Multinomial NB
Linear SGD
CNN
k-mer Transformer
0.0
0.1
0.2
0.3
0.4
0.5
0.6
Unseen-genus rejection